# YUANBO PANG

UC Berkeley EECS | Foundation Agents & Robotics
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## Education

**University of California, Berkeley** *Expected Dec 2026*
*B.S. Electrical Engineering & Computer Sciences GPA: 3.762/4.0 (in-major: 3.811)*
*Relevant coursework: CS 170, CS 168, CS 184, EECS 106A, CS 61B, CS 70*

**Stanford University (Summer Session) — CS 229: Machine Learning** *Summer 2025*

**De Anza College — Computer Science, GPA: 4.0/4.0** *2023 – 2025*

## Research Experience

**Research Assistant, Berkeley RDI / BAIR (Prof. Dawn Song)** *Oct 2025 – Present*
*Agents' Last Exam — co-first author on V1; leading two research tracks in V2*

- **Foundation Agent / 3D spatial understanding.** Building training and evaluation environments in simulation (Blender, NVIDIA Isaac Lab) where agents actively move the viewpoint to explore a scene and judge physical state — 3D understanding through exploration, not inference from a fixed 2D view.
- **Ultra-long-horizon engineering.** Constructing tasks from large-scale production software that human teams measure in months to years, with a harness that evaluates *teams* of agents — coordination, division of labor, and context handoff at horizons far beyond single-agent benchmarks.
- For V1, built task/environment pipelines, computer-use and tool interfaces, scalable GCP execution, and programmatic evaluators producing verifiable rewards for RLVR; now run ALE evaluations with frontier model teams (OpenAI, Anthropic, Meta TBD Lab, Qwen, Seed), analyzing agent trajectories and reporting failure modes back to the labs.

**Robotics Research, Prof. Masayoshi Tomizuka's Lab, UC Berkeley** *2026 – Present*
*Large-Scale Robot-Learning Data Benchmark — mentored by Zhuo Xu*

- Scoring dataset quality by downstream finetune uplift — the $\Delta$ on each model's native benchmark (LIBERO, SimplerEnv, CALVIN) — not proxy metrics; rankings stay consistent across ACT, Diffusion Policy, OpenVLA, $\pi_0$, and GR00T.
- Owning the agent-driven data layer: agents that audit annotation quality and auto-convert raw captures into the team's canonical, embodiment-tagged representation (egocentric human hand, parallel-jaw gripper, humanoid).

**URAP Research Assistant, Prof. Avideh Zakhor, UC Berkeley EECS** *Spring 2026 – Present*

- Building a high-quality UAV-to-UAV dataset and PyTorch pipelines for zero-shot detection and multi-object tracking under high-speed ego-motion; integrating optical flow with YOLO and SAM2-style baselines. Targeting CVPR submission.

## Engineering Experience

**Founder, Equile — "The dry lab for AI scientists"** *Apr 2026 – Present*

- Sole owner of the runtime and control plane: serverless hosting for agent harnesses (Claude Code, Codex, Cursor, Grok) — isolated per-session sandboxes, multi-tenant packing that cuts per-agent compute >20×, snapshot pause/resume, public SDK/CLI.

**Agentic Tech Staff, Starbot** *Nov 2025 – May 2026*

- Led the agentic stack for autonomous service robots: vision-language-action models, agentic planning and control, multi-robot coordination.
- Shipped as the CES 2026 demo system, deployed on the show floor 8 hours/day across coffee preparation, popcorn making, and front-desk reception; covered live by Japan's NHK News, demoed to 1,500+ attendees, and converted into 50+ inbound customer conversations.

**Software Engineering Intern, Geopogo** *Fall 2025*

- Shipped a production multimodal-generation workflow (React/TypeScript, FastAPI, Gemini, Firebase, CI/CD).

## Research Interests

**Foundation Agents:** 3D and environment understanding; RL systems for agent–environment interaction

**Robotics:** 3D understanding to strengthen world-action models (VLA)

## Publications

**Agents' Last Exam.**
Yiyou Sun*, Xinyang Han*, Weichen Zhang*, **Yuanbo Pang***, et al., Dawn Song. *Under review at NeurIPS 2026.* arXiv:2606.05405. (*Co-first author & core contributor; 310 authors total)
*#1 Paper of the Week, Hugging Face Daily Papers (Jun 2026).*

**ScienceIDE: Turning World's Scientific Codebase into Agent Learnable Environments.**
Hejia Geng*, Zesen Huang*, Haoyang Li, Wenbin Li, Koutian Wu, Zihan Zhou, **Yuanbo Pang**, et al., Ling Yang. arXiv:2609.19134. (AItonomy Foundation; sponsors: PhAI-Labs, Qwen)
*#2 Paper of the Day, Hugging Face Daily Papers (Sep 2026).*

**Agent's GameDev Exam.**

Zonglin Di, Yinhe Wang, **Yuanbo Pang**, Yiyou Sun, Weichen Zhang, Xinyang Han, et al., Dawn Song. *Under double-blind review at ICLR 2027.*

## Awards

HackMIT Modal Sponsor Prize (2025); Smartest AI Agent, HackMIT (2024).

## Technical Skills

**ML / RL:** PyTorch, reinforcement learning (RLVR), VLA, model finetuning & evaluation
**Vision / Robotics / Sim:** YOLO, SAM2, optical flow, Isaac Lab, Blender
**Agent / Systems:** benchmarks, automated evaluators, tool-use & MCP, Docker, GCP, Modal, FastAPI, Linux